

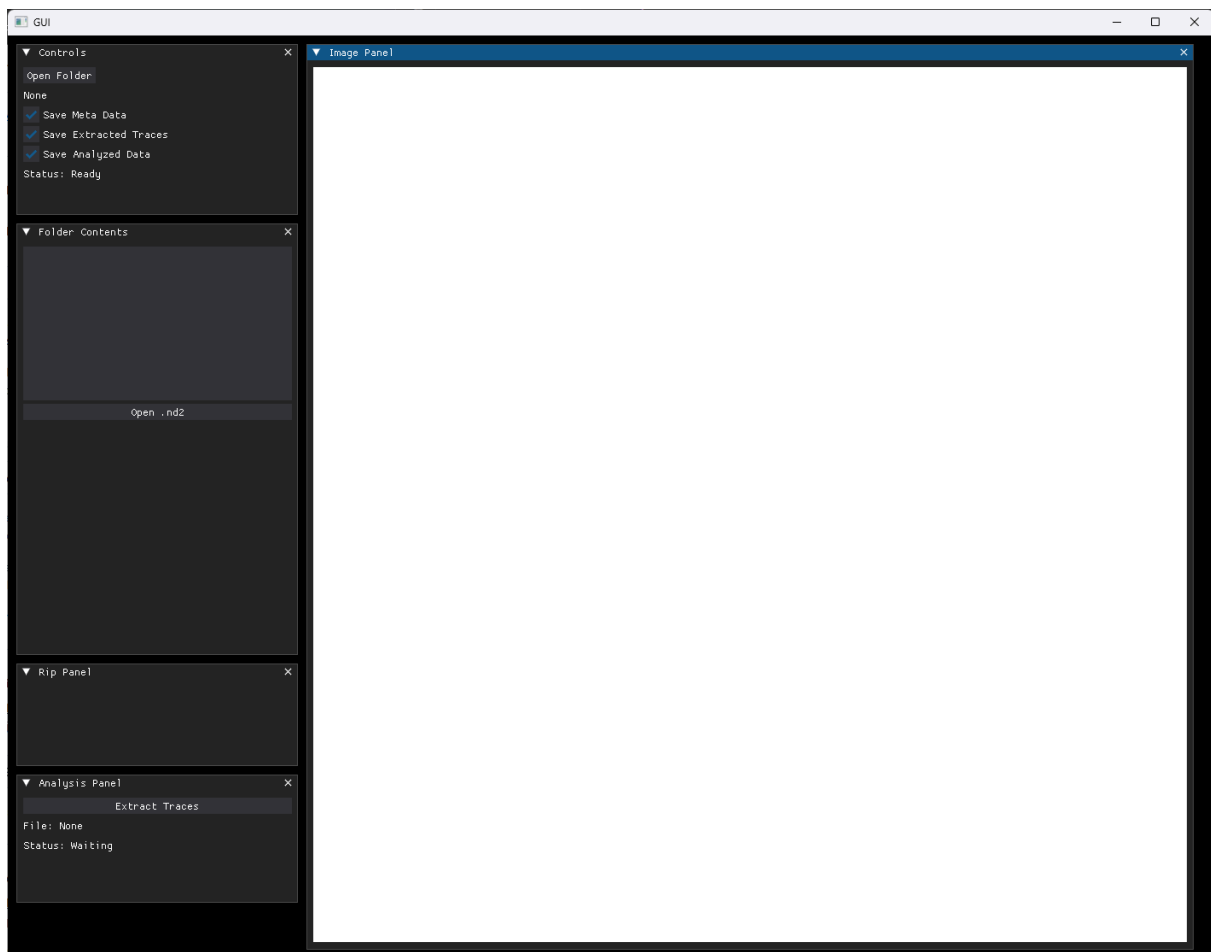
Initialization

Download Github repository: [LINK](#)

Install new conda environment: T5_env.yml

How to install a conda environment:

1. Download Anaconda ([link](#))
2. Create a new conda environment ([documentation](#))
 - a. Open Anaconda Prompt (Windows) or Terminal (Mac)
 - b. Download T5_env.yml and navigate to its directory
 - c. [Terminal] `conda env create -f T5_env.yml`
3. Activate the T5_env environment
 - a. [Terminal] `conda activate T5_env`
4. Run the GUI_imaging.py
 - a. Option 1: Open and run GUI_imaging.py in the T5_env environment
 - b. Option 2: In terminal, activate T5_env, navigate to GUI folder, and run [Terminal] `python GUI_imaging.py`



NOTICE

This version **is not connected** to DataJoint.

Before analysis, know this list of information for each of your .nd2 files

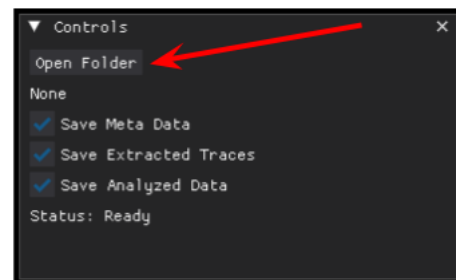
- Age (mo) of mouse
- Experimental Condition: Experimental (Flickering Light), Control (Darkness)
- Stimulus Duration (min): e.g. 0, 15, 30...

1. Ensure .nd2 file names are correct and abide by this format:
{DJID}{Eye}_{MMDDYY}_{}.nd2, where the date references the imaging date.
2. Ensure the mouse is deceased in the database.

TUTORIAL

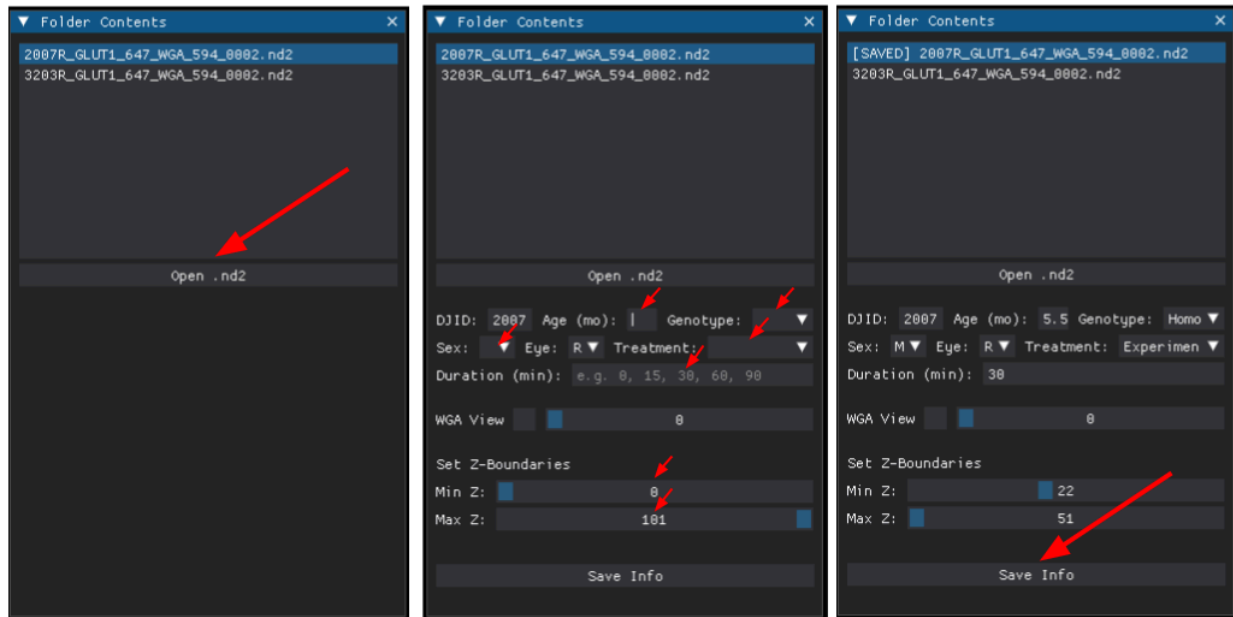
Place all your .nd2 files into a folder. Where this folder is located does not matter. At the end of the analysis, the data will be saved in a .csv format.

Click the **Open Folder** and find the folder containing the .nd2 files. All files ending with .nd2 will appear in the **Folder Contents** Panel.



Next, click **Open .nd2** on a file in the list.

- Fill in the identifying information about a file. DJID and Eye are automatically collected from the file name.
- Adjust **Set Z-Boundaries** accordingly, such that the max projection on the **ImagePanel** focuses on the cells and not artifacts above and below the RGC layer.
- Lastly, click **Save Info**. If you know if the sample contains an Inner Limiting Membrane rip, continue to **Rip Panel**.
- Continue the same process for cells.



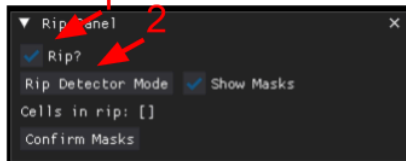
SPECIAL CASE: Cells in Rip

IGNORE IF UNECESSARY

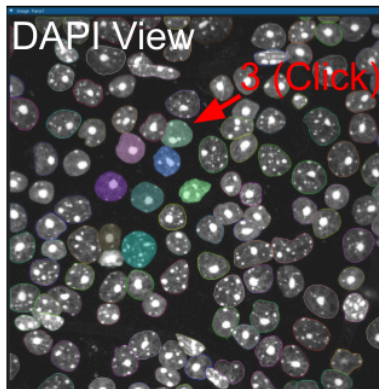
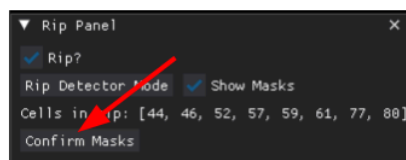
Identify cells that are exposed through rips in the Inner Limiting Membrane

- Toggle **Rip?** Checkbox
- Click **Rip Detector Mode**, which automatically toggles **Show Masks**
- Mask outlines appear such that you can select the cells in the rip by clicking on them. Toggle and scroll through the **WGA View** to better view the ILM and cells through the rip.
- The **Image Panel** updates slowly when **Show Masks** is toggled.
- When finished, click **Confirm Masks**.

Start



Finish



When all .nd2 files have their identifying information and Z-boundaries saved, see the **Analysis Panel**. Click **Extract Traces**. Only the files with [SAVED] in the **Folder Contents** Panel will be analyzed.

After the program finishes, three .csv files will appear, with the folder name as the prefix.

- metadata: identifying information inputted by the user.
- raw: raw traces per stain per cell.
- processed: analysis on the raw traces, which contains the stain integral values.

